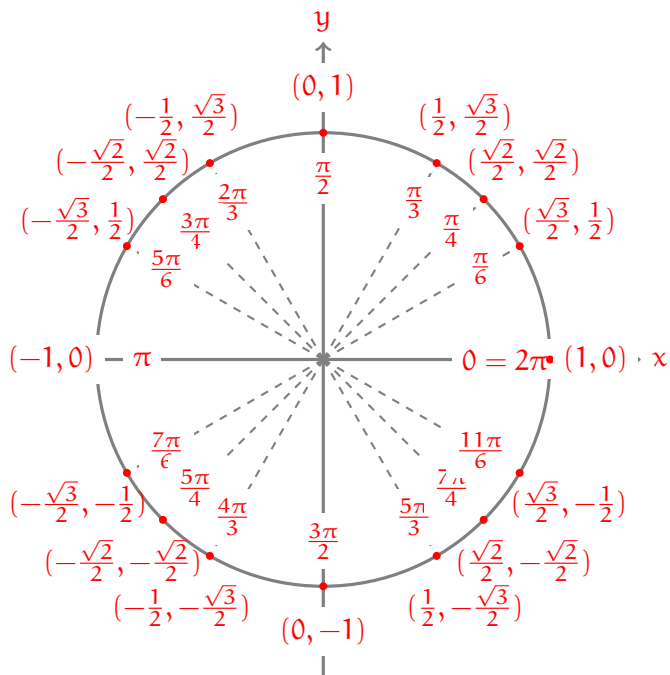


**Integral Formulas.**

- $\int \frac{1}{1+t^2} dt = \arctan t + C$
- $\int \frac{1}{\sqrt{1-t^2}} dt = \arcsin t + C$
- $\int \frac{1}{t} dt = \ln |t| + C$

Trig Identities.

- power reduction: $\cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$
 $\sin^2 \theta = \frac{1}{2} - \frac{1}{2} \cos 2\theta$
- double angle: $\sin 2\theta = 2 \sin \theta \cos \theta$
 $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$

A1 Formulas.

- $\vec{v} \cdot \vec{w} = \|\vec{v}\| \|\vec{w}\| \cos \theta$
- $\|\vec{v} \times \vec{w}\| = \|\vec{v}\| \|\vec{w}\| \sin \theta = [\text{parallelogram area}]$
- $|\vec{v} \cdot (\vec{w} \times \vec{r})| = [\text{parallelepiped volume}]$
- $\text{proj}_{\vec{v}}(\vec{w}) = \left(\frac{\vec{v} \cdot \vec{w}}{\vec{v} \cdot \vec{v}} \right) \vec{v}$
- $\text{comp}_{\vec{v}}(\vec{w}) = \frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\|}$

A2 Formulas.

- distance from point **B** to plane $\mathcal{P}$ with normal $\vec{n}$:
 $\frac{|\vec{AB} \cdot \vec{n}|}{\|\vec{n}\|}$ where **A** is on $\mathcal{P}$
- distance from point **B** to line ℓ with direction $\vec{v}$:
 $\frac{\|\vec{AB} \times \vec{v}\|}{\|\vec{v}\|}$ where **A** is on ℓ

A3 Formulas. Standard equations.

- paraboloid: $z = x^2 + y^2$ 
- saddle: $z = x^2 - y^2$ 
- 1-sheeted hyperboloid: $x^2 + y^2 - z^2 = 1$ 
- 2-sheeted hyperboloid: $-x^2 - y^2 + z^2 = 1$ 
- ellipsoid—sphere if uniformly scaled:
 $x^2 + y^2 + z^2 = 1$ 
- double-cone: $z^2 = x^2 + y^2$ 

A4 Formulas.

- tangent plane to $z = f(x, y)$ at $(a, b, f(a, b))$ is:
 $z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$

A5 Formulas.

- given unit direction $\vec{u}$, the directional derivative:
 $D_{\vec{u}}f(P) = \vec{\nabla}f(P) \cdot \vec{u} = \|\vec{\nabla}f(P)\| \cos \theta$
- tangent plane to level set $F(x, y, z) = C$ at **P** is:
 $\vec{\nabla}F(P) \cdot \vec{x} = \vec{\nabla}F(P) \cdot \vec{p}$

B1 Formulas.

- If **P** is a critical point of $f(x, y)$ and:
 $D = f_{xx}(P)f_{yy}(P) - (f_{xy}(P))^2$
 $T = f_{xx}(P) + f_{yy}(P)$
- then:
 $D < 0 \implies$ saddle
 $D > 0$ and $T > 0 \implies$ local minimizer
 $D > 0$ and $T < 0 \implies$ local maximizer

B2 Formulas.

- An extremizer **P** of f subject to $g = C$ satisfies:
 $\vec{\nabla}f(P) = \lambda \vec{\nabla}g(P)$ or $\vec{\nabla}g(P) = \vec{0}$

B5 Formulas.

- polar and cylindrical
 - $r^2 = x^2 + y^2$ and $\tan \theta = \frac{y}{x}$
 - $x = r \cos \theta$ and $y = r \sin \theta$
 - $dA = r dr d\theta$ and $dV = r dz dr d\theta$

C1 Formulas.

- spherical
 - $\rho^2 = x^2 + y^2 + z^2$ and $\tan \phi = \frac{r}{z}$
 - $r = \rho \sin \phi$ and $z = \rho \cos \phi$
 - $x = \rho \sin \phi \cos \theta$ and $y = \rho \sin \phi \sin \theta$
 - $dV = \rho^2 \sin \phi \, d\rho d\phi d\theta$

C2 Formulas.

- arclength $= \int_C 1 \, ds = \int_a^b \|\vec{r}'(t)\| \, dt$
- scalar line integral:
 - $\int_C f \, ds$ where $ds = \|\vec{r}'(t)\| \, dt$
- vector line integral:
 - $\int_C \vec{F} \cdot d\vec{r}$ where $d\vec{r} = \vec{r}'(t) \, dt$
 - or $\int_C P \, dx + Q \, dy + R \, dz$ where:
 $\langle dx, dy, dz \rangle = \langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \rangle \, dt$

C3 Formulas.

- fundamental theorem of line integrals:
 - given $\vec{F} = \nabla f$ and path $\vec{r}(t)$, $a \leq t \leq b$, then:
 $\int_{\vec{r}} \vec{F} \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$
- conservative vector fields are irrotational ($\text{curl} = \mathbf{0}$)
- irrot. + simply-conn. domain $\implies$ conservative
- $\text{curl } \vec{F} = \nabla \times \vec{F}$
- 2D vector fields $\vec{F} = \langle P, Q \rangle$ have: $\text{curl } \vec{F} = (Q_x - P_y)\vec{k}$

C4 Formulas.

- parametrized surface $\vec{R}(u, v)$
 - infinitesimal normal: $d\vec{S} = \pm \vec{R}_u \times \vec{R}_v \, dx dy$
 - infinitesimal surf. area: $dS = \|\vec{R}_u \times \vec{R}_v\| \, dx dy$
- shortcut: parameters x and y
 - $d\vec{S} = \pm \langle -\frac{\partial z}{\partial x}, -\frac{\partial z}{\partial y}, 1 \rangle \, dx dy$
- shortcut: surface $z = f(r)$ with parameters r, θ
 - $d\vec{S} = \pm \langle -\frac{\partial z}{\partial r} \cdot r \cos \theta, -\frac{\partial z}{\partial r} \cdot r \sin \theta, r \rangle \, dr d\theta$
 - $dS = r \sqrt{\left[\frac{\partial z}{\partial r}\right]^2 + 1} \, dr d\theta$
- shortcut: sphere $\rho = R$ with parameters ϕ, θ
 - $d\vec{S} = (R \sin \phi) \langle x, y, z \rangle \, d\phi d\theta$
 - $dS = R^2 \sin \phi \, d\phi d\theta$

C5 Formulas.

- Green's Theorem:
 - $C = \partial D$, oriented so D on left, then:
 $\oint_C P \, dx + Q \, dy = \iint_D Q_x - P_y \, dA$
 - Stokes's Theorem:
 - $C = \partial S$, oriented by righthand rule, then:
 $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot d\vec{S}$
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